

The Sharp Rank and Squared-Mass Cost of Spectral Prefix Realization

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Abstract

Exact finite-prefix realizations exist by RH-316. We determine their optimal complexity scale. If $s = q_*/q$, then both the least rank and the least squared spectral mass needed to match through order N are $\Theta(s^N)$. The lower bound is a one-moment consequence of the modulus cap; the recursive packet construction gives the matching upper bound. The result concerns synthetic spectra and does not identify an actual noisy rank.

1 Universal lower bounds

Let $\mathcal{S}$ be a finite multiset in $|\mu| \leq q$, with rank r and squared mass $M_2 = \sum |\mu|^2$. If its N th moment equals a_N , then

$$|a_N| \leq r q^N, \quad |a_N| \leq M_2 q^{N-2}.$$

Since RH-268 proves $a_N/q_*^N \rightarrow 1$, both r and M_2 are bounded below by positive constant multiples of s^N , where $s = q_*/q > 1$.

2 Recursive upper bound

Theorem 2.1. *For the deterministic anchor prefix through order N , the RH-316 recursion can be chosen so that*

$$r_N \leq C_1 s^N, \quad M_{2,N} \leq C_2 s^N.$$

Consequently the minimal rank and minimal squared mass are both $\Theta(s^N)$.

Proof. Let $\mathcal{P}_e$ be the packet inserted at order e , and write b_e and r_e for its squared mass and rank, taking both to be zero if no packet is needed. The all-order envelope gives $|a_d| \leq A q_*^d$ for a fixed A .

The decisive point is the divisor sparsity of RH-315: an e -packet contributes to the d th moment only when $e \mid d$. Hence, for $d \geq 2$, the residual before the d th correction obeys

$$|w_d| \leq A q_*^d + q^{d-2} \sum_{\substack{e \mid d \\ e < d}} b_e.$$

Indeed each contributing packet satisfies $|p_d(\mathcal{P}_e)| \leq q^{d-2} b_e$. With $L_d = \lceil |w_d|/(dq^d) \rceil$ when $w_d \neq 0$, one has

$$r_d = d L_d \leq A s^d + q^{-2} \sum_{\substack{e \mid d \\ e < d}} b_e + d,$$

and, because the new packet lies in $|\mu| \leq q$,

$$b_d \leq q^2 r_d \leq A q^2 s^d + \sum_{\substack{e \mid d \\ e < d}} b_e + q^2 d.$$

Every proper divisor of d is at most $d/2$. Thus, if $b_e \leq Cs^e$ for $e < d$, the divisor sum is at most $Cds^{d/2}$. Since $ds^{-d/2} \rightarrow 0$, enlarging C to cover finitely many initial orders closes the induction and gives $b_d = O(s^d)$. The rank display then also gives $r_d = O(s^d)$. Finally

$$M_{2,N} = \sum_{d \leq N} b_d = O(s^N), \quad r_N = \sum_{d \leq N} r_d = O(s^N).$$

The preceding one-moment inequalities give the matching lower bounds. □

Numerically,

$$s = 1.4017504517095518\dots, \quad \log s = 0.3377217782684642\dots$$

Remark 2.2. The theorem is an extremal law inside the finite normal spectral class. It does not assert that the actual noisy complement has minimal rank, obeys this mass asymptotic, or matches the deterministic prefix. Gates A–E remain false/open.